SYSTEM DESIGN CHEAT SHEET

50 System Design Patterns

Every Engineer Should Know

Memorize architectures and you learn one answer.

Learn the patterns and you can design any system.

Swipe for all 50 →

Primary-Replica

WHAT IT DOES

One primary database handles writes. Replicas copy the writes and serve reads. If the primary fails, a replica is promoted.

REAL EXAMPLE

PostgreSQL streaming replication, MySQL read replicas in AWS RDS.

WHEN TO USE

Read-heavy systems: social feeds, product catalogs, user profiles.

Sharding

WHAT IT DOES

Split data across multiple database servers. A shard key decides which shard holds which data.

REAL EXAMPLE

Discord shards messages by guild_id across thousands of Cassandra nodes.

WHEN TO USE

When one database cannot handle the write volume or storage.

Consistent Hashing

WHAT IT DOES

Place servers and keys on a virtual ring. Adding or removing a node only remaps a small slice of keys.

REAL EXAMPLE

DynamoDB, Cassandra, and Redis Cluster all use consistent hashing for partitioning.

WHEN TO USE

Distributed caches and databases where nodes are added or removed often.

Write-Ahead Log (WAL)

WHAT IT DOES

Append every operation to a sequential log before writing to storage. On crash, replay the log to recover.

REAL EXAMPLE

Postgres WAL, MySQL binlog, Cassandra commit log.

WHEN TO USE

Foundation of every durable database and the source of replication streams.

Event Sourcing

WHAT IT DOES

Store the full sequence of events instead of the current state. Current state is derived by replaying events.

REAL EXAMPLE

Banking ledgers, Git commit history, Kafka-based order systems.

WHEN TO USE

Audit-heavy domains: finance, compliance, anywhere you need to rebuild past states.

CQRS

WHAT IT DOES

Separate the write model (commands) from the read model (queries). Each is optimized independently.

REAL EXAMPLE

Orders written to Postgres, served to users from a denormalized Elasticsearch index.

WHEN TO USE

Systems with very different read and write patterns or shapes.

Cache-Aside

WHAT IT DOES

App checks the cache. On a miss, it reads the database, returns the result, and populates the cache.

REAL EXAMPLE

Redis in front of MySQL, the default caching pattern at most companies.

WHEN TO USE

The go-to caching strategy when reads vastly outnumber writes.

Write-Through

WHAT IT DOES

Every write hits the cache and the database at the same time. Cache is always fresh.

REAL EXAMPLE

User profile updates that must be visible immediately after save.

WHEN TO USE

Systems where read-after-write consistency is required.

Write-Behind

WHAT IT DOES

Writes go to the cache first. The cache asynchronously flushes batches to the database.

REAL EXAMPLE

Gaming leaderboards, analytics counters, view-count incrementers.

WHEN TO USE

Write-heavy paths where short-lived inconsistency is acceptable.

Read-Through

WHAT IT DOES

The cache itself fetches from the database on a miss. The app only ever talks to the cache.

REAL EXAMPLE

AWS DAX in front of DynamoDB, NCache, Apache Ignite.

WHEN TO USE

When you want all caching logic centralized in the cache layer.

Cache Stampede Prevention

WHAT IT DOES

When a hot key expires, thousands of requests can hit the DB at once. Coalesce, lock, or refresh early.

REAL EXAMPLE

Celebrity profile pages, trending product pages on Black Friday.

WHEN TO USE

Any cache with popular keys and a hard TTL.

Request-Response

WHAT IT DOES

Client sends a request and blocks until it gets a response. The simplest communication model.

REAL EXAMPLE

REST APIs, gRPC unary calls, GraphQL queries.

WHEN TO USE

User-facing APIs, lookups, anything that needs an immediate answer.

Message Queue

WHAT IT DOES

Producer drops a message on a queue. Consumer reads and processes it on its own schedule.

REAL EXAMPLE

Sending welcome emails via SQS, processing uploads via RabbitMQ.

WHEN TO USE

Decoupling services when the caller does not need an immediate result.

Publish-Subscribe

WHAT IT DOES

Publisher emits to a topic. Every subscriber gets a copy. One-to-many fan-out, not one-to-one delivery.

REAL EXAMPLE

OrderCreated event triggering inventory, notifications, and analytics at once.

WHEN TO USE

Event-driven systems where multiple services react to the same event.

Event-Driven Architecture

WHAT IT DOES

Services emit and react to events instead of calling each other directly. Workflows happen through event chains.

REAL EXAMPLE

Uber rides, Netflix microservices, modern e-commerce backends.

WHEN TO USE

Loosely coupled microservices that need to evolve independently.

Webhooks

WHAT IT DOES

Server POSTs to a client-provided URL when an event happens. The opposite of polling.

REAL EXAMPLE

Stripe payment events, GitHub push events, Slack slash command callbacks.

WHEN TO USE

Third-party integrations and any push-style notification flow.

Server-Sent Events

WHAT IT DOES

A long-lived HTTP connection where the server pushes events to the client. One-way only.

REAL EXAMPLE

Live sports tickers, news feeds, build status dashboards.

WHEN TO USE

Real-time updates where the client only needs to receive, not send.

Bidirectional Streaming

WHAT IT DOES

A persistent connection where both sides send messages at any time. WebSockets and gRPC streaming.

REAL EXAMPLE

Slack and WhatsApp chat, Google Docs collaboration, multiplayer games.

WHEN TO USE

Real-time, two-way communication with low latency.

Circuit Breaker

WHAT IT DOES

Stop calling a failing service. Return a fallback instantly. Periodically test for recovery.

REAL EXAMPLE

Netflix Hystrix, Resilience4j, Istio outlier detection.

WHEN TO USE

Any service that depends on a flaky downstream dependency.

Retry with Backoff

WHAT IT DOES

On failure, retry after increasing delays (1s, 2s, 4s, 8s) plus random jitter to avoid herds.

REAL EXAMPLE

AWS SDKs, Stripe client libraries, Kubernetes controllers.

WHEN TO USE

Any transient failure: network blips, timeouts, brief overload.

Bulkhead

WHAT IT DOES

Isolate workloads into separate resource pools. If one pool drowns, the others stay afloat.

REAL EXAMPLE

Separate thread pools per downstream service, per-tenant connection limits.

WHEN TO USE

Multi-tenant systems and mixed batch plus real-time workloads.

Timeout

WHAT IT DOES

Cap how long any external call can take. If it goes over, abort and return an error or fallback.

REAL EXAMPLE

HTTP client timeouts, gRPC deadlines, database query timeouts.

WHEN TO USE

Every external call in every distributed system. No exceptions.

Idempotency

WHAT IT DOES

Design write operations so running them twice gives the same result as running them once.

REAL EXAMPLE

Stripe Idempotency-Key header, message dedup in Kafka consumers.

WHEN TO USE

Every write API. Network failures make duplicate retries inevitable.

Dead Letter Queue

WHAT IT DOES

After N failed retries, move the message to a separate queue instead of blocking the main one.

REAL EXAMPLE

AWS SQS DLQ, Kafka error topics, RabbitMQ x-dead-letter-exchange.

WHEN TO USE

Every message-driven system. Stops one poison message from killing the pipeline.

Graceful Degradation

WHAT IT DOES

When non-critical pieces fail, serve a reduced experience instead of a full outage.

REAL EXAMPLE

Show trending items when personalization is down. Skip images when CDN is slow.

WHEN TO USE

Any user-facing system with subsystems of varying criticality.

Horizontal Scaling

WHAT IT DOES

Add more machines to handle more traffic. Ten servers handle ten times the load of one.

REAL EXAMPLE

Adding more EC2 instances behind a load balancer, more pods in Kubernetes.

WHEN TO USE

Stateless services: web servers, API servers, background workers.

Vertical Scaling

WHAT IT DOES

Upgrade to a bigger machine. More CPU, more RAM, faster disks.

REAL EXAMPLE

Resizing an RDS instance from db.m5.large to db.m5.24xlarge.

WHEN TO USE

Databases and single-threaded workloads. A quick fix before going horizontal.

Load Balancing

WHAT IT DOES

Spread incoming requests across servers using round-robin, least-connections, or IP hash.

REAL EXAMPLE

AWS ALB, NGINX, HAProxy, Envoy.

WHEN TO USE

Every horizontally scaled service. The first thing you add at scale.

Auto-Scaling

WHAT IT DOES

Add or remove instances automatically based on traffic, CPU, or queue depth.

REAL EXAMPLE

AWS Auto Scaling Groups, Kubernetes HPA, GCP Managed Instance Groups.

WHEN TO USE

Cloud systems with bursty traffic: flash sales, live events, ad campaigns.

Connection Pooling

WHAT IT DOES

Keep a pool of reusable database connections instead of opening a new one per request.

REAL EXAMPLE

HikariCP for JVM, pgbouncer for Postgres, mysql-pool for Node.

WHEN TO USE

Every application that talks to a database, at any scale.

MapReduce

WHAT IT DOES

Split data across machines (map), process each chunk in parallel, combine the results (reduce).

REAL EXAMPLE

Hadoop, Google indexing, Spark batch jobs.

WHEN TO USE

Terabyte-scale batch processing: analytics, ETL, log aggregation.

Stream Processing

WHAT IT DOES

Process events as they arrive, one by one, with sub-second latency.

REAL EXAMPLE

Kafka Streams, Apache Flink, Spark Structured Streaming.

WHEN TO USE

Real-time analytics, fraud detection, live recommendations.

Lambda Architecture

WHAT IT DOES

Run batch and stream pipelines in parallel. Batch gives accuracy, stream gives speed. Serving layer merges both.

REAL EXAMPLE

Twitter timeline, LinkedIn analytics platforms historically.

WHEN TO USE

Analytics platforms that need both real-time and batch-accurate views.

Change Data Capture

WHAT IT DOES

Capture inserts, updates, and deletes from a database as a stream. Downstream systems react.

REAL EXAMPLE

Debezium, AWS DMS, Postgres logical replication to Kafka.

WHEN TO USE

Sync a search index, a cache, or a data warehouse with the primary database.

API Gateway

WHAT IT DOES

Single entry point that routes, authenticates, rate-limits, and transforms requests for all backend services.

REAL EXAMPLE

AWS API Gateway, Kong, Apigee, Netflix Zuul.

WHEN TO USE

Every microservices architecture. One front door instead of 20.

Backend for Frontend

WHAT IT DOES

A dedicated API layer per client type. Mobile gets a lean response, web gets a richer one.

REAL EXAMPLE

Netflix mobile vs TV vs web BFFs. SoundCloud per-platform BFFs.

WHEN TO USE

Apps where mobile, web, and internal clients have very different needs.

Rate Limiting

WHAT IT DOES

Cap how many requests a client can make per time window. Token bucket, fixed window, or sliding window.

REAL EXAMPLE

GitHub API limits, Stripe per-key limits, Twitter API tiers.

WHEN TO USE

Every public API. Stops abuse, buggy clients, and runaway loops.

Cursor Pagination

WHAT IT DOES

Return pages using an opaque cursor pointing to the last item. Client passes the cursor for the next page.

REAL EXAMPLE

Twitter, Facebook, and Slack APIs all use cursor pagination.

WHEN TO USE

Any list endpoint that can return thousands of items: feeds, search, logs.

API Versioning

WHAT IT DOES

Run multiple API versions in parallel so existing clients keep working when the API evolves.

REAL EXAMPLE

/v1/users vs /v2/users, or version headers like Accept: vnd.api+json;v=2.

WHEN TO USE

Any public API with third-party clients you cannot force to upgrade.

CDN

WHAT IT DOES

Replicate static content to edge servers around the world. Users hit the closest edge.

REAL EXAMPLE

Cloudflare, Akamai, AWS CloudFront, Fastly.

WHEN TO USE

Any system serving static assets or with a global user base.

Reverse Proxy

WHAT IT DOES

Server that sits in front of backends. Handles SSL, compression, caching, and routing.

REAL EXAMPLE

NGINX, HAProxy, Envoy, Traefik.

WHEN TO USE

Every production web service. Offloads cross-cutting concerns from app code.

Service Mesh

WHAT IT DOES

Infrastructure layer for service-to-service traffic. Sidecars handle retries, mTLS, observability.

REAL EXAMPLE

Istio, Linkerd, AWS App Mesh, Consul Connect.

WHEN TO USE

Large microservices fleets where adding reliability per-service is unsustainable.

Sidecar

WHAT IT DOES

A helper process deployed alongside the main service. Handles logging, config, or networking.

REAL EXAMPLE

Envoy sidecar in Istio, Fluentd log shipper, Vault Agent sidecar.

WHEN TO USE

Adding capabilities to a service without changing its code.

Two-Phase Commit

WHAT IT DOES

Coordinator asks all participants to prepare. If all agree, commit. If any refuses, roll back.

REAL EXAMPLE

Distributed transactions in XA-compliant databases, classic ATM cross-bank transfers.

WHEN TO USE

Strict atomicity across services. Use sparingly. It blocks under failure.

Saga Pattern

WHAT IT DOES

A chain of local transactions. If a step fails, run compensating transactions to undo the prior ones.

REAL EXAMPLE

E-commerce checkout: reserve inventory, charge card, ship order, each with a rollback.

WHEN TO USE

Long-running distributed workflows where 2PC is too slow or fragile.

Quorum

WHAT IT DOES

With N replicas, require W writes and R reads where W + R > N. Guarantees the latest write is read.

REAL EXAMPLE

DynamoDB, Cassandra, etcd, ZooKeeper all use quorum-based consistency.

WHEN TO USE

Distributed databases that offer tunable consistency per operation.

Vector Clocks

WHAT IT DOES

A vector of logical timestamps, one per node. Tracks causality and surfaces concurrent updates.

REAL EXAMPLE

Amazon Dynamo paper, Riak, version vectors in distributed key-value stores.

WHEN TO USE

Conflict detection in eventually consistent systems.

Health Check Endpoint

WHAT IT DOES

Every service exposes /health returning its status. Load balancers poll it to route only to healthy instances.

REAL EXAMPLE

Kubernetes liveness and readiness probes, AWS ELB target group health checks.

WHEN TO USE

Every production service. Without it, load balancers serve crashed pods.

Distributed Tracing

WHAT IT DOES

Track a request as it flows across services. Each service adds a span. The full trace shows the path and latency.

REAL EXAMPLE

Jaeger, Zipkin, AWS X-Ray, Datadog APM, OpenTelemetry.

WHEN TO USE

Any microservices system where you need to find latency or failure sources.

Canary Deployment

WHAT IT DOES

Roll out a new version to a small percent of traffic. Watch metrics. Expand if healthy, roll back if not.

REAL EXAMPLE

Spinnaker canaries, Argo Rollouts, LaunchDarkly progressive delivery.

WHEN TO USE

Every production deployment. Catches bugs that testing misses.

YOU MADE IT TO 50

That is the vocabulary.
Now combine them.

If you can explain each pattern in 2 to 3 sentences with a concrete example, you have the vocabulary for any system design conversation.

WANT THE FULL 90-MINUTE GUIDE?

Every pattern, every trade-off, every interview-tested example.

Read the full breakdown on DesignGurus.io

Link in the comments →

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